

Grade 6 Accelerated
Unit 9
Lesson 1 Homework

Name _____
Date _____
Period _____

For questions 1–7, complete the table below.

	Data Description	Examples	Data Classification
1.	Number of siblings a student has		
2.	Breed of cat in a pet store	Tabby, Persian, Siamese, Bengal, Sphynx	
3.			categorical
4.			numerical
5.	Shoe size of basketball players	10, 10.5, 11, 11.5, 12, 13	
6.	Color of shoes of basketball players		

7. _____ data, or quantitative data, is a type of data that is measurable. It is collected in the form of numbers.

For questions 8–10, classify each as Categorical or Numerical.

8. Color of shirts people are wearing in a movie theater.

9. Make of car each teacher at your school drives.

10. Weight of each pet you own.

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Lesson 2 Homework

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For questions 1–7, determine whether each question is statistical or not statistical.

1. How many cousins do you have?
2. What country is North of the United States of America?
3. What time does Math class start?
4. What shoe size do you wear?
5. What is your favorite food?
6. What is the capital of Florida?

7. The question "What is the largest Alligator length ever recorded?" is not statistical. Use the definition to justify why it is not a statistical question.
8. Decide if the question, "How tall are you?" is statistical or non-statistical. Explain your decision.
9. Create an example of a statistical question.
10. Create a counterexample that is NOT a statistical question.

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Lesson 3 Homework

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Jillian took a survey of her class to find out how many sports teams each student joined throughout the last year. Use the data in the table for questions 1–3.

Number of Sports Teams Joined									
0	1	6	2	4	3	0	2	4	2

1. Determine the mean of the data.
2. Interpret the mean in terms of the context.
3. What is the median number of sports teams students joined?

For one week Ruth kept track of how long it took for her in minutes to walk to school. Use the data for questions 4–5.

Day of the week	Total Minutes
Monday	25
Tuesday	22
Wednesday	26
Thursday	18
Friday	20

4. Find the mean.
5. Circle the data piece that is closest to the mean amount of time she spent walking to school.

6. To calculate the mean, _____ all the values in the data set and then _____ by the total number of data values.

Florida is one of the largest producers of citrus fruits in the United States. The table below shows the amount in millions of citrus fruits produced by season. Use the data for questions 7–8.

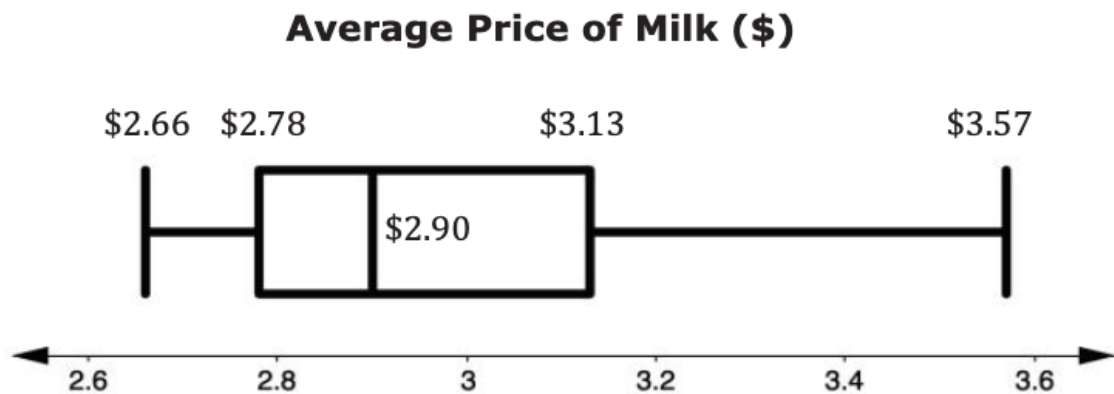
Season	Amount of Fruit Produced (in millions)
2001 – 2002	12.824
2002 – 2003	11.206
2003 – 2004	13.045
2004 – 2005	7.597
2005 – 2006	7.832
2006 – 2007	7.236
2007 – 2008	9.119

7. Find the mean of the data.
8. Interpret the mean amount of fruit produced based on the data.
9. The Tampa Bay Rays had 6 outfielders in their 2021 season. Find the mean of their ages.

Pitchers Name	Pitchers Age
Randy Arozarena	27
Kevin Kiermaier	31
Josh Lowe	24
Manuel Margot	27
Austin Meadows	26
Brett Phillips	27

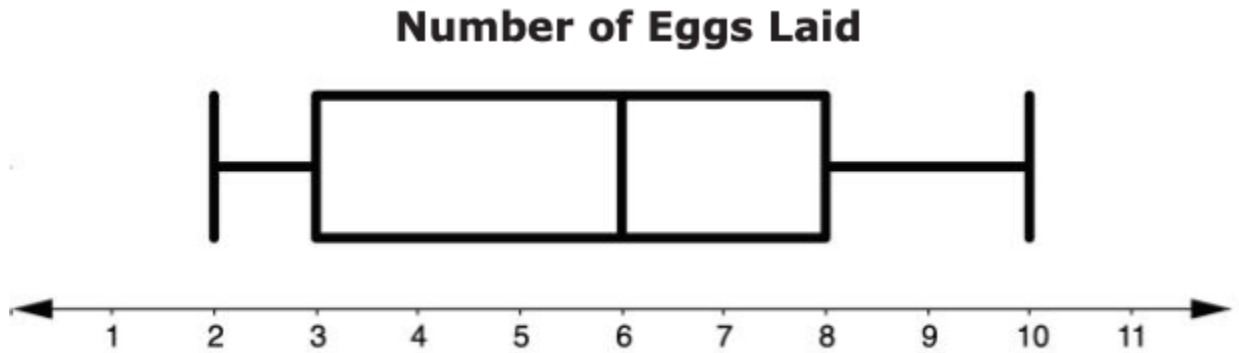
10. Circle the data pieces that are closest to the mean.

The average price of milk, in dollars (\$), for each month for 7 years was recorded and used to create the box plot shown. Use the box plot for questions 1-6.



1. The minimum price is _____, while the maximum price is _____.
2. The median price is _____.
3. The upper quartile is _____.
4. The lower quartile is _____.
5. The range is _____.
6. The interquartile range is _____. The IQR was calculated by subtracting _____ from _____.

A farmer recorded the number of eggs her farm full of hens laid in one week. The box plot represents the number of eggs laid by each hen during that week. Use the box plot for questions 7-10.



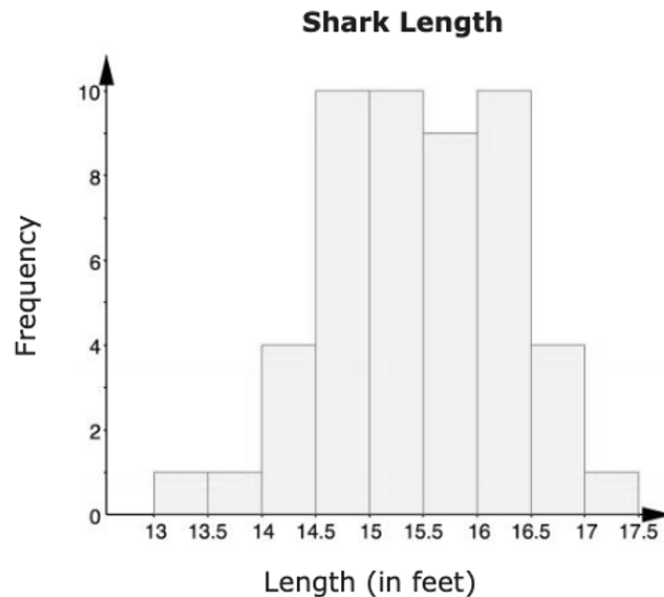
7. Write a description of the spread and distribution of the number of eggs laid.

8. What percentage of hens laid 2 or more eggs?

9. What percentage of hens laid between 6 and 10 eggs?

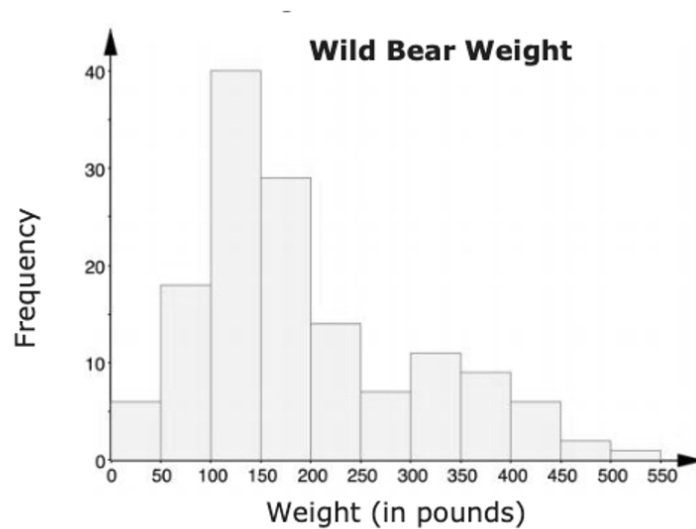
10. What percentage of hens laid between 2 and 8 eggs?

The histogram summarizes the lengths, in feet (ft), of a group of adult female sharks.



1. How many sharks are between 16.5 and 17 feet?
2. How many sharks measured are less than 15 feet?
3. What is the smallest possible length of a shark in this data set?
4. Are there more female sharks measuring less than 15.5 feet or more than 15.5 feet?
5. If sharks younger than adult age were added to the sample, hypothesize what would happen to the spread of the data?

In a study of wild bears, researchers measured the weights, in pounds (lb.) of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



6. Is it possible that the sample contains a bear that weighs 50 pounds?
7. More than half of the bears weigh less than _____ pounds in this study.
8. Mark an "M" above the bar that contains the approximate median.
9. Mark an "L" above the bar that contains the bear that weighs the least.
10. Which single range of weights has the most bears?

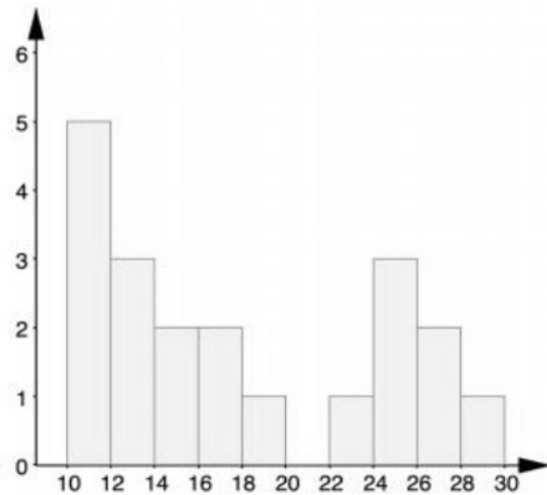
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Lesson 6 Homework

Name _____
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1. A _____ skewed graph has a longer trail of data to the left and the peak of the data on the right.

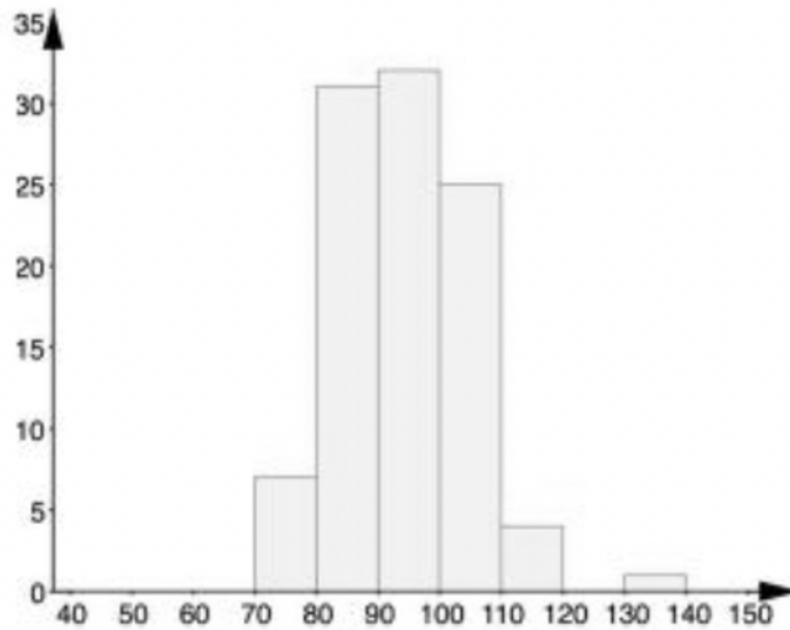
2. The shape of the histogram is

- ☐ symmetric.
- ☐ left skewed.
- ☐ right skewed.



3. If the data value _____ were added to the histogram above, it would be considered an outlier.
4. Sketch a histogram whose shape can be described as symmetric.
5. Sketch a histogram whose shape can be described as left skewed.
6. If an outlier is added to a data set that is far less than the median, which direction will the graph be skewed?

Use the histogram for questions 7–10.

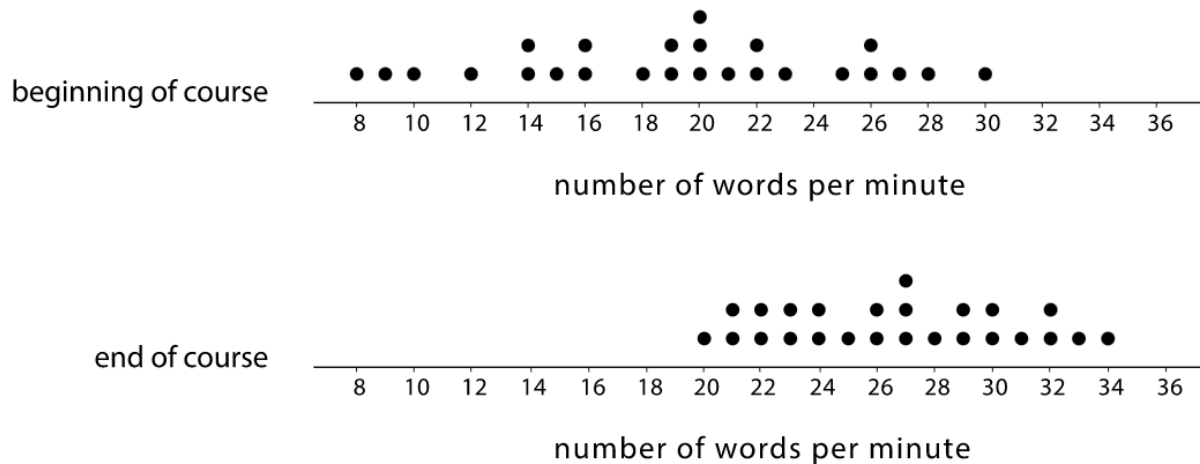


7. The shape of the histogram can be described as
- ☐ symmetric.
 - ☐ left skewed.
 - ☐ right skewed.
8. Is there a gap in the histogram? If so, where?
9. Is there a cluster in the histogram? If so, where?
10. If the data value _____ were added to the histogram above, it would be considered an outlier.

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Lesson 7 Homework

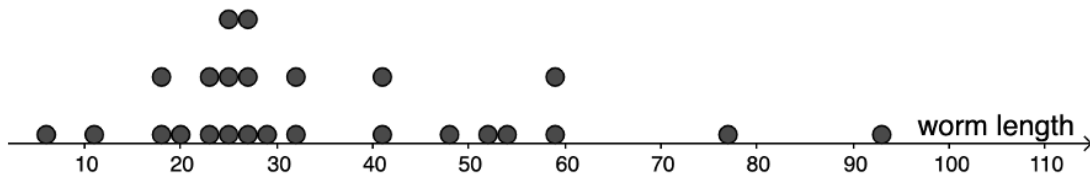
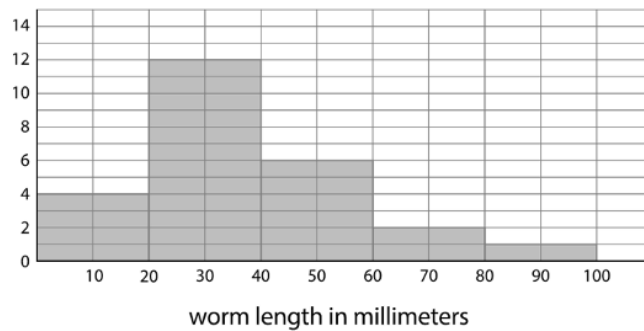
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The teacher recorded the number of words that her students could type per minute at the beginning of a course and again at the end. The two dot plots show the two data sets.



1. How many students were in the course?
2. Determine the mean number of words typed per minute at the beginning of the course.
3. Determine the mean number of words typed per minute at the end of the course.
4. What percentage of students typed 22 words per minute or more at the beginning of the course versus the end of the course?
5. Describe the spread and variability of the data from the beginning of the course to the end of the course.

An earthworm farmer set up several containers of a certain species of earthworms so that he could learn about their lengths. The lengths of the earthworms provide information about their ages. The dot plot and histogram display the measure of the lengths of earthworms in one of the containers. Each length was measured in millimeters.

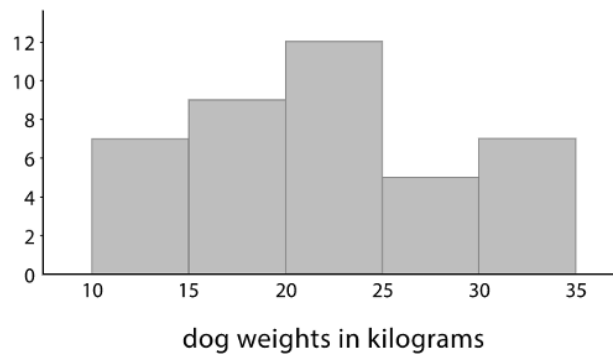
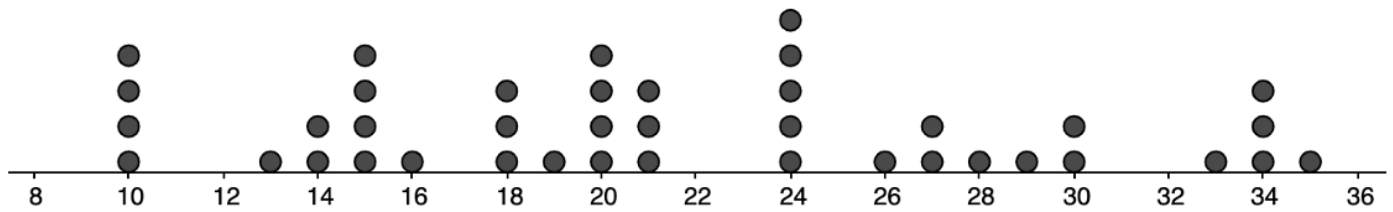


6. To determine the range, which data display should be used and why?
7. What do you notice about the shape of the distribution?
8. How many earthworms were measured?
9. Which data display can be said to determine how many earthworms measured 25 millimeters?
10. How many earthworms measured 25 millimeters?

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Lesson 8 Homework

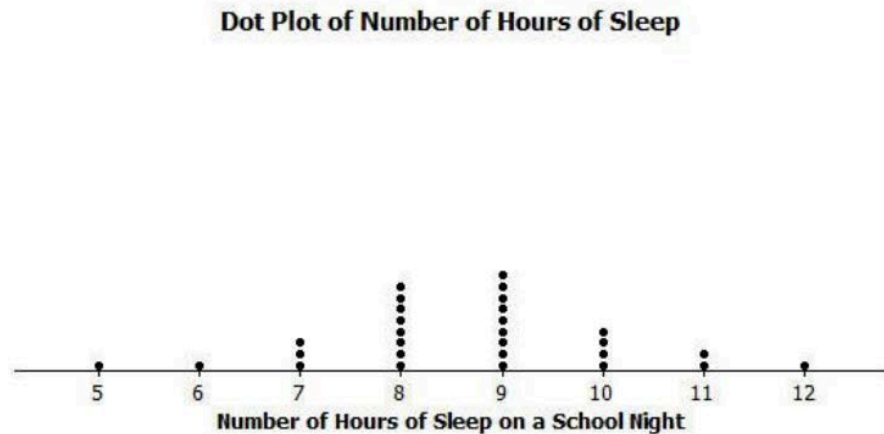
Name _____
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A histogram and dot plot show the weights, in pounds, of 40 dogs at a dog show.



1. Which data display can be used to determine the mean weight?
2. Find the mean weight. Interpret the mean.
3. Which data display can be used to determine the median weight?
4. Find the median weight. Interpret the median.

Rocky is a sixth grader at Roosevelt Middle School and investigated the number of hours of sleep sixth-grade students get on school nights. He surveyed 29 sixth graders and organized his data on the dot plot.



5. Describe the spread and shape of the distribution.
6. Find the mean number of hours slept by 6th graders.
7. Interpret the mean.
8. Find the median.
9. Interpret the median.
10. Suppose Rocky surveyed 10 more students and they reported the following (in hours); 9, 10, 9, 9, 12, 12, 12, 8, 10, 7. If these 10 data points were added to the dot plot, explain how would the mean and median change.

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The ages of 28 people in a movie theater are shown.

6, 6, 7, 7, 7, 8, 8, 8, 8, 9, 9, 9, 10, 10, 11, 13, 14, 18, 19, 25, 27, 28, 28, 33, 37, 40, 45, 67

1. Determine the five-number summary for the data set.

Minimum: _____

First Quartile: _____

Median: _____

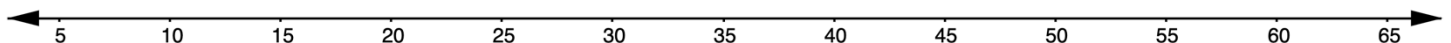
Third Quartile: _____

Maximum: _____

2. What is the range of the data?

3. What is the interquartile range of the data?

4. Create a box plot of the ages at the movie theater.



A bakery's data on the number of muffins sold each day for the last 14 days are shown.

32	41	65	58	31	38	46	39	38	62	60	34	31	44
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5. Determine the five-number summary for the data set.

6. Create a box plot of the bakery's daily muffin sales.



7. Explain how you determine the scale to use for the number line of your box plot.

LeBron James's average points per game (PPG) for each regular season that he's been in the NBA are shown.

Season	03-04	04-05	05-06	06-07	07-08	08-09	09-10	10-11	11-12	12-13
PPG	20.9	27.2	31.4	27.3	30.0	28.4	29.7	26.7	27.1	26.8
Season	13-14	14-15	15-16	16-17	17-18	18-19	19-20	20-21	21-22	
PPG	27.1	25.3	25.3	26.4	27.5	27.4	25.3	25.0	29.0	

8. Determine the five-number summary for the data set.

9. Create a box plot of LeBron James's average points per game.



10. What is the interquartile range?

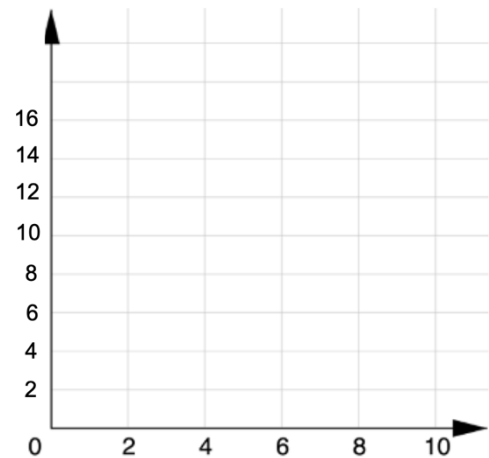
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Luna collected data on the number of pets her classmates have. Her data is shown.

0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 2, 2, 3, 3, 7

1. Create a frequency chart of the data. 2. Create a histogram of the data.

Number of Pets	Frequency

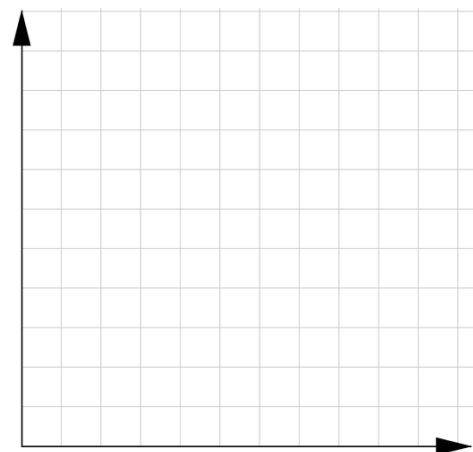


Marcus is training for a half-marathon. His daily miles run for the last 20 days are shown.

3.5	3	2.5	3.5	6	0	3.5	4	2.5	2
3	6.5	0	4	4	3.5	4.5	2	7.5	0

3. Create a frequency chart of the data. 4. Create a histogram of the data.

Number of Miles	Frequency



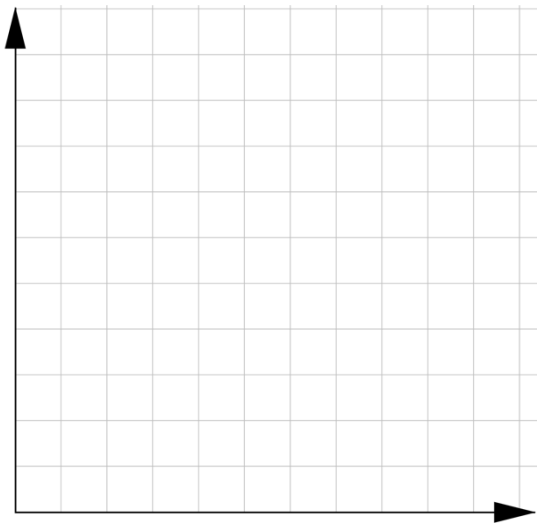
5. Explain how you determined the scale for the axes on your histogram in question 4.

The heights, in inches, of twelve players on a college basketball team are shown.

71	73	70	79	68	70	73	72	68	74	72	71
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6. Create a frequency chart of the data.
7. Create a histogram of the data.

Height (in.)	Frequency

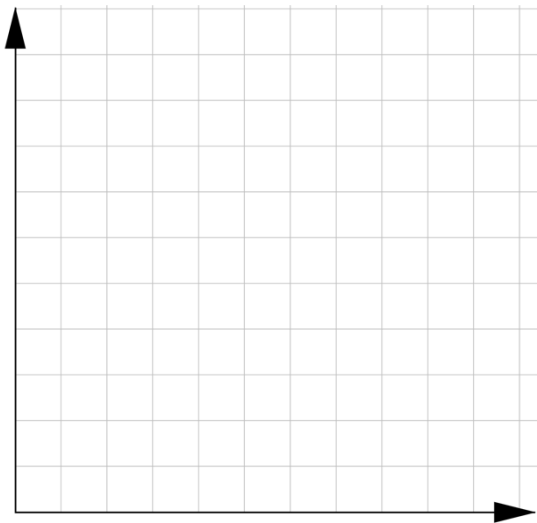


The heights, in inches, of 12 players on a sixth-grade basketball team are shown.

64	73	60	67	68	58	72	53	62	65	70	61
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8. Create a frequency chart of the data.
9. Create a histogram of the data.

Height (in.)	Frequency



10. How are your histograms for questions 7 and 9 different?